

KEVIN RUSAGARA

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EDUCATION

Yale University

New Haven, CT | Aug 2023 – May 2027

Bachelor of Science in Computer Science

Relevant Coursework: Algorithms, Computer Architecture, Object-Oriented Programming, Distributed Systems, Compilers, Artificial Intelligence, Machine Learning

SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL

Frameworks & Tools: FastAPI, Flask, React, Spring Boot, AWS, Docker, Git, Linux, PyTorch, TensorFlow, NumPy, FAISS, LangChain

EXPERIENCE

Dispatch Energy | Software Engineer Intern

New York, NY | Summer 2026

- Engineered MCP-based automation tools for analysts, reducing recurring project diligence and market research workflows from 3 hours to 3 minutes, a 98.3% decrease in manual analysis time.
- Built an AI contract compliance agent that validates proposed actions, emails, and workflow changes against entity-specific contract clauses, flags potential breaches, and routes outputs for human review in under 5 minutes.
- Developed an ISO-level electricity market prediction tool that integrated cleaned market datasets with analyst inputs to support forecasting, validation, and investment decision-making across multiple power markets.

Yale University & NSF | Software Engineer

New Haven, CT | Aug 2025 – Present

- Building software infrastructure for a hyperspectral imaging ecosystem that helps researchers analyze materials and biological samples beyond the visible spectrum.
- Developing React-based dashboard tools for visualizing, transforming, and managing high-dimensional spectral image datasets.
- Building data conversion pipelines that standardize complex hyperspectral image files into formats researchers can process, compare, and share.

UL Solutions | Software Engineer Intern

Fremont, CA | Jun 2025 – Aug 2025

- Developed Python automation tools to control RF test equipment and orchestrate SAR measurement workflows, reducing manual setup time and improving test repeatability.
- Partnered with RF and embedded systems engineers to debug test workflows, document compliance processes, and build scripts for firmware version tracking and reporting.

Awesomity Lab | Backend Engineer Intern

Kigali, Rwanda | Jun 2024 – Aug 2024

- Developed the backend for Provigator, a production bug-tracking system used internally to manage and resolve 320+ software issues.
- Implemented RESTful APIs with FastAPI and SQLite, supporting full CRUD operations for users, projects, clients, and client-project relationships, and integrated Google OAuth authentication

PROJECTS

AI Agent Identity & RAG Gateway

Python, FastAPI, JWT, FAISS, LangChain

- Built a backend system for secure AI agent access with JWT authentication, role-based permissions, and API endpoints for controlled query execution.
- Implemented a RAG pipeline using FAISS vector search and LangChain to retrieve relevant documents across private knowledge sources.
- Designed permissioned REST APIs for tool execution, document retrieval, and traceable access across AI-assisted workflows.

High Street Health(Home-Care Operations Platform)

PostgreSQL, SQLAlchemy, Next.js, React, AWS

- Built a HIPAA-aligned clinical operations platform used in a live home-health pilot, serving clinician, patient, and mobile field workflows across the patient lifecycle.
- Designed FastAPI and async SQLAlchemy APIs for intake, scheduling, vitals, versioned clinical notes, escalation alerts, soft deletes, Alembic migrations, and PHI audit logging.
- Implemented C-CDA discharge XML ingestion with vendor-schema mapping, S3 KMS storage, source-traceable intake prefill, Cognito auth, and AWS deployment across shared web/mobile clients.